

ARE WE PLAYING GOD?

Critics of modern science say that when we interfere with nature, we play God. Scientific breakthroughs often arrive with a warning that we are straying into forbidden territory. Scientists argue that we change the course of nature every time we take antibiotics, breed animals for food, or build a dam across a river. They feel that as long as we improve the lives of humans, the end justifies the means. So, is there a problem with playing God? Someone has to pull the strings . . .

IT'S SHOWTIME!

GENETIC MODIFICATION

In genetic modification, scientists alter or add to DNA in order to change the genes of an organism, such as a plant or animal. In one type of modification, somatic gene therapy, doctors swap bad genes for functioning ones to treat disease. It is difficult and risky, but in the future, we might be able to use this therapy to wipe out genetic diseases.

DESIGNER BABIES

Germ-line gene therapy takes place in the sperm and egg cells, altering the traits a child is born with. Eventually, this may help parents prevent genetic disorders or hereditary diseases affecting their unborn children. Germ-line therapy is illegal in many countries, however, and its opponents fear that people will use it someday to create a flawless "designer baby" that is smarter, more talented, and more good-looking than nature intended.

LIFE EXTENSION

Some people view aging and death as medical conditions that may be delayed through the use of genetic engineering. Humans could alter their genes to stop muscle deterioration or even to regrow limbs and other organs. Using genetic engineering to create "super people" who live long lives is more controversial than using it to help people who are sick.

MODIFIED MENU

Plants can be genetically modified, too. The first GM foods were made to resist viruses and herbicides. The next generation was created to be hardier and more nutritious. In the future, crops may contain edible drugs and vaccines. Through GM, we can increase food production and decrease the chemicals used to kill pests. Critics of GM foods say that we don't yet know what eating them might do to us and to the planet's ecosystem.